

Caregiver Failed Extubation Rate and Patient Survival:

Working Document — May 2026

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This is a working document summarizing analyses completed to date and identifying specific analyses needed to answer questions raised by Andrew and Ryan. Blue boxes are action items. All analyses use MIMIC-IV v3.1 (Beth Israel Deaconess Medical Center, 2008–2022).

1 Cohort and Data

1.1 Invasive vs. non-invasive ventilation

There is an indicator variable in MIMIC-IV distinguishing invasive (endotracheal) intubation from non-invasive ventilation (CPAP/BiPAP). I confirmed that the pipeline filters to **InvasiveVent** only. The other categories are **SupplementalOxygen**, **NonInvasiveVent**, **HFNC**, **Tracheostomy**.

1.2 Cohort sizes and covariate comparison

Stage	N patients	Lost
InvasiveVent ≥ 4 h in MIMIC-IV v3.1	27,729	
of whom: explicit extubation documented	18,958	
of whom: extubation inferred from vent record	8,771	
<i>Filters applied to both groups:</i>		
Died in hospital without extubation		−6,050
Transitioned to tracheostomy		−1,523
No computable ventilation duration		−8,068
Incomplete covariates or non-ICU unit		−68
patient_cohort (last extubation per patient)	12,662	
Inferred events or missing caregiver assignment		−8,280
explicit_extubations (analytical cohort)	4,382	
Unique caregivers	271	
ICU units	9	

Table 1: Cohort flow from all invasively ventilated patients in MIMIC-IV to the analytical cohort. The **patient_cohort** (N=12,662) is used for descriptive analyses; the **explicit_extubations** cohort (N=4,382) is used for all caregiver-level analyses. The largest single exclusion is patients with no computable ventilation duration (−8,068), reflecting a documentation alignment gap between the `procedureevents` and `ventilation` derived tables.

	All patients	Explicit cohort
N	12,662	4,382
Age, mean (SD)	62.7 (16.2)	63.1 (15.7)
Female, %	40.7	38.5
Charlson, mean (SD)	5.57 (3.25)	5.45 (3.25)
SOFA, mean (SD)	6.91 (3.86)	6.39 (3.54)
Norepinephrine, mean (SD)	0.17 (0.23)	0.14 (0.19)
Vent hours, mean (SD)	94.3 (134.1)	64.6 (94.7)
Medical-respiratory intubation, %	74.7	64.4
Circulatory primary diagnosis, %	32.9	40.2
12-month survival, %	54.8	67.7
Hospital mortality, %	27.3	12.4

Table 2: Comparison of the full patient cohort (explicit + inferred extubation events) vs. the analytical cohort (explicit events with genuine caregiver assignment). The analytical cohort is systematically less sick: lower SOFA, shorter ventilation, lower mortality. Restriction to explicit events introduces selection bias toward patients with better-documented, shorter ICU stays.

A note on caregiver attribution: `caregiver_id` in MIMIC-IV records the *documenting* user, not necessarily the provider who performed the extubation. This is a known limitation of the dataset.

Non-ICU units excluded: Five unit labels were identified as non-ICU settings and excluded from the analytical cohort: PACU ($N = 19$), generic “Intensive Care Unit (ICU)” ($N = 12$), Surgery/Vascular/Intermediate ($N = 76$), and Medicine ($N = 11$). The Surgery/Vascular and Medicine units show unusual profiles (50% COVID diagnoses, vent hours >220 h) suggesting possible COVID-era overflow wards rather than standard ICU admissions. Final analytical cohort: $N = 4,382$.

1.3 Terminal extubation

The current analysis uses death within 3 days as the criterion, flagging 193 patients (4.5% of the analytical cohort). A sensitivity analysis excluding these cases is reported in Section 4.3. If some other threshold is the preferred clinical criterion, the sensitivity analysis can be repeated at any threshold.

2 ICU Unit Breakdown

2.1 Unit clinical profiles

The nine recognized ICU units fall into four clinically distinct groups which are used for a stratified Propensity Score Matching:

Post-cardiac surgery (CVICU, $N = 2,585$). Oldest patients (mean 66.5y), 84% circulatory diagnoses, short ventilation (55h mean), lowest FE rate (1.7%), best survival (80%). Extubation is a near-routine step in surgical recovery. This is the group where caregiver FE rate predicts survival.

Medical critical care (MICU + MICU/SICU, $N = 4,537$). Highest SOFA (8.4 and 7.6), longest ventilation (118h and 111h), 88%+ medical-respiratory intubation. MICU dominated by infectious (28%) and respiratory (20%) diagnoses; MICU/SICU by neoplasms (16%), infectious

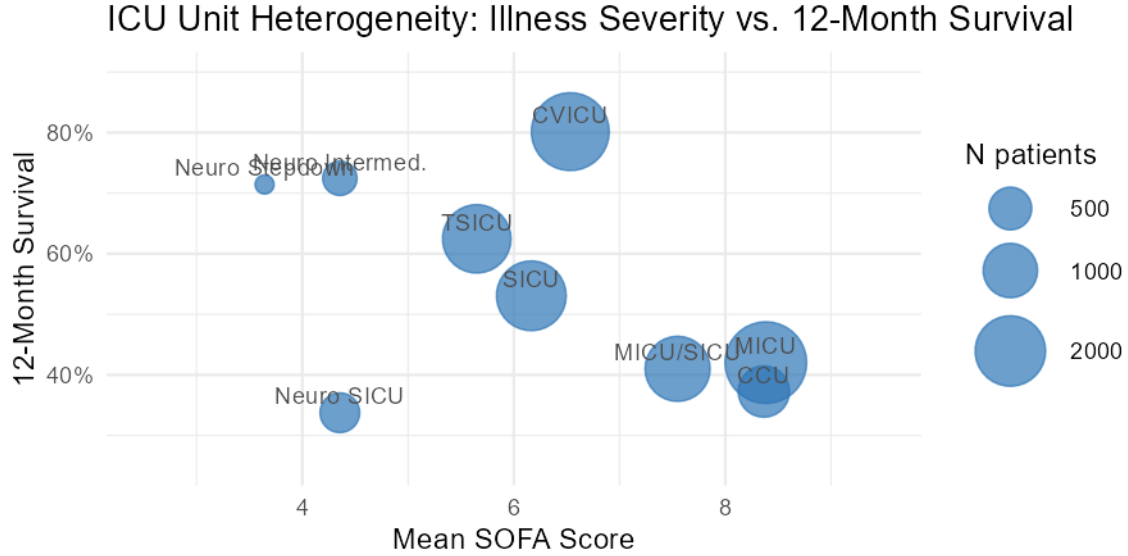


Figure 1: ICU unit heterogeneity: mean SOFA score vs. 12-month survival. Bubble size proportional to patient count.

Unit	N	Age mean (SD)	SOFA mean (SD)	Charlson mean (SD)	Vent h median [IQR]	12-mo surv %
CVICU	2,585	66.5 (12.5)	6.5 (3.3)	5.4 (2.9)	14 [6–45]	80.2
TSICU	1,854	58.7 (19.2)	5.7 (3.5)	4.3 (3.3)	44 [15–122]	62.5
SICU	1,947	62.7 (15.9)	6.2 (4.0)	5.6 (3.0)	42 [17–123]	53.1
MICU	2,915	61.3 (16.5)	8.4 (4.0)	6.0 (3.3)	64 [23–154]	42.0
MICU/SICU	1,622	61.8 (16.9)	7.6 (3.8)	6.1 (3.5)	54 [18–142]	41.0
CCU	851	67.6 (13.9)	8.4 (3.5)	7.0 (2.9)	50 [20–114]	37.3
Neuro SICU	403	64.7 (16.3)	4.4 (3.0)	5.7 (3.2)	33 [15–82]	33.7
Neuro Intermed.	265	58.9 (17.0)	4.4 (2.9)	4.9 (3.0)	35 [17–81]	72.5
Neuro Stepdown	98	58.6 (16.3)	3.6 (2.2)	4.4 (3.0)	25 [12–66]	71.4

Table 3: Patient profiles by ICU unit (patient_cohort, N=12,662; explicit + inferred events). Units ordered by 12-month survival. Age, SOFA, and Charlson reported as mean (SD); ventilation hours as median [IQR], reflecting the right-skewed distribution. See Table 5 for unit-specific FE rates in the analytical cohort. There is no separate CTICU label in MIMIC-IV. The cardiac surgery recovery unit appears exclusively as “Cardiac Vascular Intensive Care Unit (CVICU).”

(27%), and respiratory (19%). 12-month survival 42% and 41%. No caregiver FE rate signal detected.

Surgical and trauma (SICU + TSICU, N = 3,801). TSICU: youngest (58.7y), lowest Charlson (4.3), 46% injury/poisoning diagnoses, 63% survival. SICU: more heterogeneous (17.9% digestive, 15.5% injury, 11.2% neoplasms), 53% survival. Short-term mortality reversal observed in this group (see Section 4.1).

Cardiac (CCU, N = 851). Oldest alongside CVICU (67.6y), highest Charlson (7.0), highest norepinephrine use (0.25), 64% circulatory diagnoses. 37% 12-month survival despite only 40.8% hospital mortality — substantial post-discharge mortality. Excluded from stratified PSM due to

small caregiver count in the explicit analytical cohort (4 caregivers with $n > 10$ events), insufficient for quartile comparison.

Neuro units ($N = 766$). Neuro SICU: 52.6% hospital mortality despite low SOFA (4.4) — neurological injury severity is poorly captured by SOFA. Neuro Intermediate and Stepdown have better survival (72% and 71%) but are step-down rather than full ICU settings.

3 Caregiver Profiles

The analytical cohort includes 132 caregivers with ≥ 5 patients in the explicit sample, spanning 6 ICU units. 73 of 132 caregivers (55%) are primarily CVICU. Table 4 summarizes caregiver-level statistics by primary unit.

	CVICU	MICU	TSICU	SICU	MICU/SICU	CCU
N caregivers	73	30	12	8	6	3
<i>FE rate</i>						
Median	0%	4.4%	6.0%	1.1%	7.7%	0%
IQR	0–3%	2–7.5%	5.6–8.3%	0–3.3%	3.9–10.4%	0–0%
Max	25%	12.8%	17.6%	6.7%	20%	0%
<i>Volume (total extubations)</i>						
Median	89	172	36	35	65	23
IQR	42–161	36–458	20–60	28–224	34–297	18–24
Max	713	983	75	387	519	25
<i>Patients in analytical sample</i>						
Median	16	24	7	9	14	6
12-mo survival, mean	77.8%	57.5%	67.1%	56.0%	52.5%	47.2%

Table 4: Caregiver profiles by primary ICU unit. Volume = total explicit extubation events in MIMIC-IV (all units, all years). Patients in analytical sample = patients in `explicit_extubations` attributed to this caregiver. CVICU caregivers have the lowest FE rates (median 0%) and best patient survival (mean 77.8%). MICU caregivers have the highest volume (median 172) reflecting the larger and busier unit. CCU has only 3 caregivers with $n \geq 5$ patients in the analytical sample.

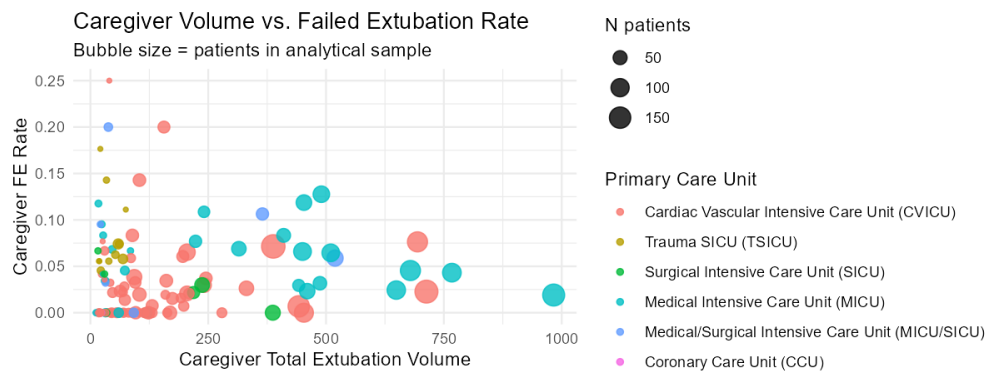


Figure 2: Caregiver total extubation volume vs. failed extubation rate. Bubble size = patients in analytical sample.

4 Propensity Score Matching (PSM) and Inverse Probability Weighting (IPW)

The propensity score model for each unit group includes: age, sex, Charlson comorbidity index, first-day SOFA score, norepinephrine equivalent dose, ventilation hours, intubation type (surgical / medical-respiratory / medical-non-respiratory), and 21 binary diagnosis flags derived from ICD-10 categories. MIMIC-IV does not include a pre-computed APACHE score.

Figure 3 shows covariate balance before and after matching in the CVICU group. All standardized mean differences fall below 0.10 after matching.

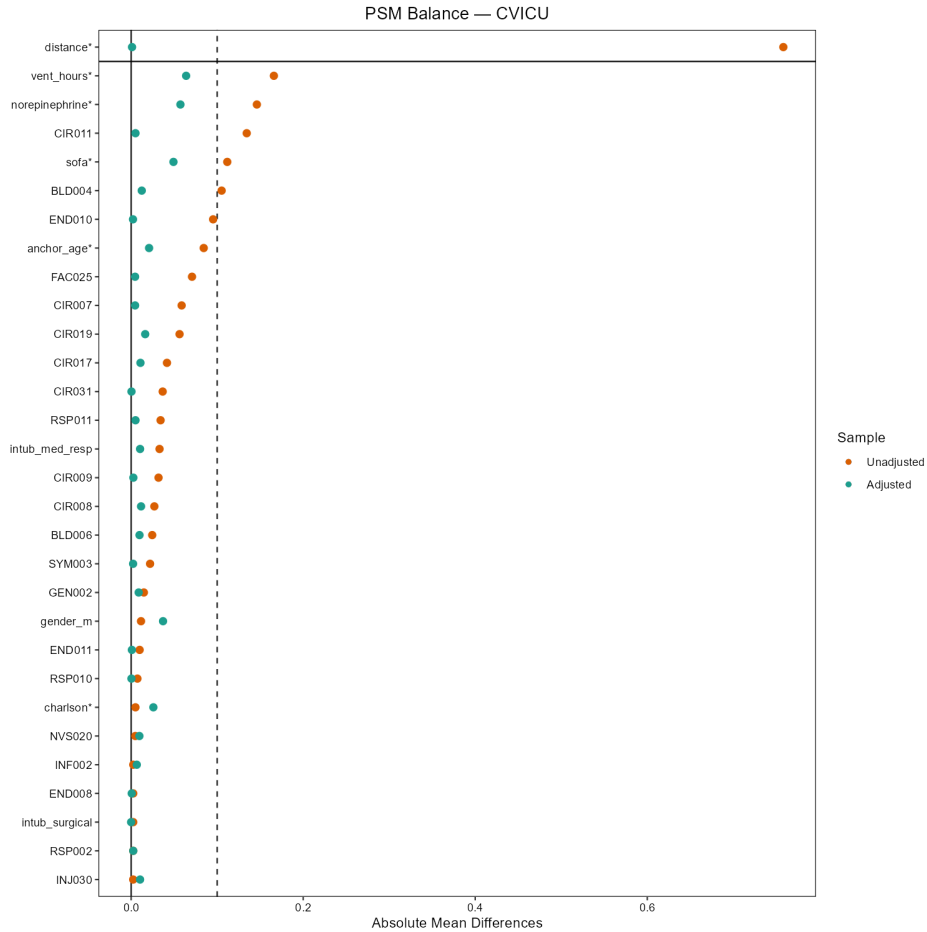


Figure 3: Covariate balance before and after PSM — CVICU.

The analysis compares patients of top-quartile FE rate caregivers to bottom-quartile FE rate caregivers, within each unit group. For PSM the matching is 1:1 nearest-neighbor within a 0.2 SD caliper on the logit propensity score, with exact matching on unit within multi-unit groups. IPW uses the Average Treatment Effect (ATE) estimand as a sensitivity check. In this context the "treatment" is being a patient of a high-FE-rate caregiver vs a low-FE-rate caregiver. The ATE answers: if we randomly assigned every patient in the population to either a high-FE or low-FE caregiver, what would the average difference in survival be?

Group	N matched	Surv (high FE)	Surv (low FE)	PSM OR (adj)	<i>p</i>	IPW OR
CVICU	797	84.4%	90.1%	0.581 (0.356–0.937)	0.027	0.656
Medical	607	49.4%	49.2%	1.091 (0.754–1.584)	0.645	1.157
Surgical/Trauma	524	72.9%	68.2%	1.202 (0.767–1.889)	0.423	1.171

Table 5: PSM and IPW results by unit group. The odds ratio (OR) is the ratio of the odds of 12-month survival for patients of high-FE-rate caregivers vs. low-FE-rate caregivers. $OR < 1$ indicates better survival for patients of low-FE-rate caregivers.

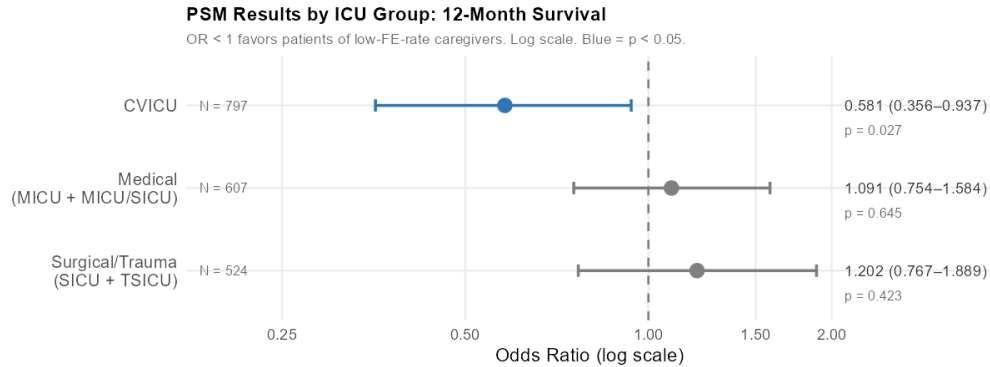


Figure 4: Propensity Score Matching by ICU group for 12 month survival as a function of caregiver FE rate.

4.1 Secondary outcomes

Group	Hosp. mortality OR	<i>p</i>	30-day mortality OR	<i>p</i>
CVICU	1.89 (0.89–4.28)	0.109	1.69 (0.67–4.58)	0.276
Medical	1.09 (0.73–1.61)	0.683	1.13 (0.76–1.67)	0.544
Surgical/Trauma	0.557 (0.321–0.949)	0.034	0.546 (0.304–0.958)	0.038

Table 6: Secondary outcomes (mortality endpoints).

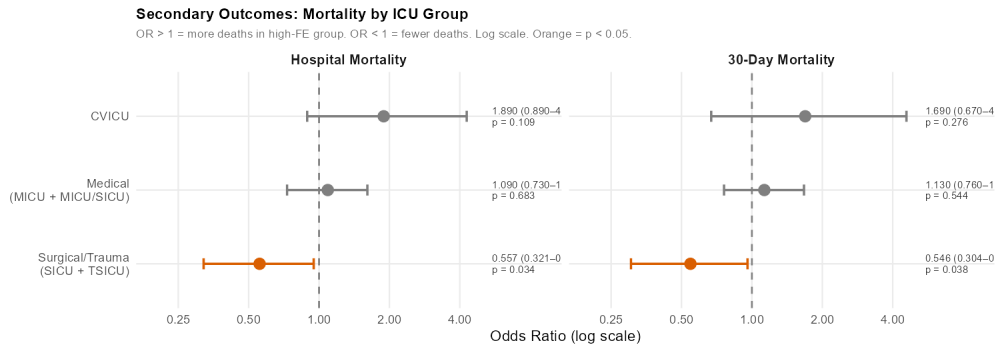


Figure 5: Propensity Score Matching by ICU group for hospital and 30-day mortality as a function of caregiver FE rate.

These secondary endpoints are mortality outcomes rather than survival outcomes. This means the direction of the odds ratio is reversed here: an OR > 1 means higher mortality for patients of high-FE-rate caregivers, whereas the survival table uses OR < 1 to mean better survival for low-FE-rate caregivers. CVICU: no acute mortality signal despite a 12-month survival effect. Surgical/Trauma: short-term mortality is lower in the high-FE-rate group.

TODO - survival → mortality Redo the analysis with 12 month mortality so that the three outcomes match direction.

4.2 Tracheostomy trajectory (Surgical/Trauma)

Of 169 failed extubations in SICU/TSICU, only 3 (1.8%) led to tracheostomy within 30 days. The tracheostomy pathway does not explain the short-term mortality reversal in this group. The reversal remains unexplained and is noted as a limitation.

4.3 Comfort extubation sensitivity

Excluding likely comfort extubations (in-hospital death within 3 days of extubation; $N = 11$ in CVICU, 0.7% of the CVICU group):

	PSM OR	p	IPW OR	p
CVICU (all)	0.581	0.027	0.656	0.005
CVICU (comfort excluded)	0.619	0.055	0.673	0.010

The CVICU signal is not driven by comfort-care confounding.

4.4 Volume threshold sensitivity

Restricting to caregivers with more patients in the analytical sample:

Min. sample patients	N caregivers	r (FE rate $\sim$ survival)	p
≥ 5	134	-0.288	< 0.001
≥ 10	88	-0.255	0.017
≥ 20	52	-0.354	0.011

Partial correlation (FE rate $\sim$ 12-month survival, controlling for caregiver volume) strengthens with higher volume thresholds. Volume alone does not predict survival once FE rate is controlled ($r = -0.109$, $p = 0.211$).

5 Functional form: survival floor

Fitting $P(\text{survival}) = b + (1 - b)e^{-k \cdot \text{FE rate}}$ at the caregiver level ($N = 78$ caregivers with FE rate > 0):

Parameter	Estimate	95% CI
b (survival floor)	0.624	0.586–0.658
k (decay rate)	77.5	52.7–122.4

Approximately 62% of patients in this cohort survive 12 months regardless of caregiver FE rate. The remaining 38% represent outcomes potentially sensitive to caregiver extubation practice. The decay is steep: most of the survival difference occurs in the 0–8% FE rate range.

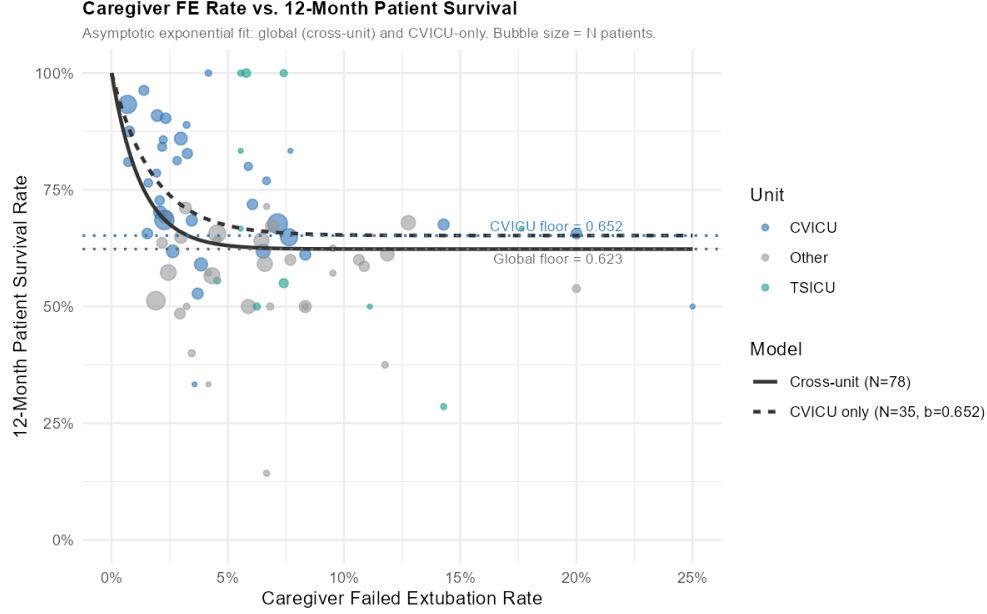


Figure 6: Caregiver rates and 12-month survival

6 Mixed effects models

Model	Estimate	95% CI	$p / P(<0)$
glmmTMB (caregiver_fe_rate)	−0.554	–	$p = 0.539$
rstanarm (caregiver_fe_scaled)	−0.022	(−0.102, 0.059)	70.7%

Random effect variances: caregiver_id ≈ 0 (both models); first_careunit = 0.271 (glmmTMB), posterior mean 0.43 (rstanarm).

The global mixed effects model is null ($p = 0.539$, $P(< 0) = 70.7\%$). The unit-level random effect absorbs the survival differences across units, leaving the caregiver FE rate fixed effect with nothing to explain at the global level. This is consistent with the PSM finding that the signal is CVICU-specific.

7 Deep Learning Models

A Python pipeline trains three models on the same patient-level covariates used in the R analyses (70/15/15 train/val/test split, seed 237): L2-regularized logistic regression, gradient-boosted trees (GBM), and a multilayer perceptron (MLP) with entity embeddings for ICU unit, intubation type, primary diagnosis, and sex. These are predictive benchmarks, not causal models. They ask whether flexible nonlinear interactions improve discrimination and whether caregiver FE rate contributes to prediction once the network can exploit richer feature interactions.

7.1 Predictive performance

Model	AUC-ROC	AUC-PR
Logistic regression	0.806	0.935
Gradient boosting	0.828	0.943
MLP (val, best ep. 16)	≈ 0.825	—

The MLP matches gradient boosting closely (val AUC ≈ 0.825 at early-stopping epoch 16), indicating no deep nonlinear structure beyond what the tree ensemble already captures at this sample size ($N \approx 4,300$). All models substantially outperform the survival-rate baseline (0.77 precision at zero recall), consistent with the R regression models showing that SOFA, Charlson, and ICU unit predict outcomes well.

7.2 SHAP feature importance

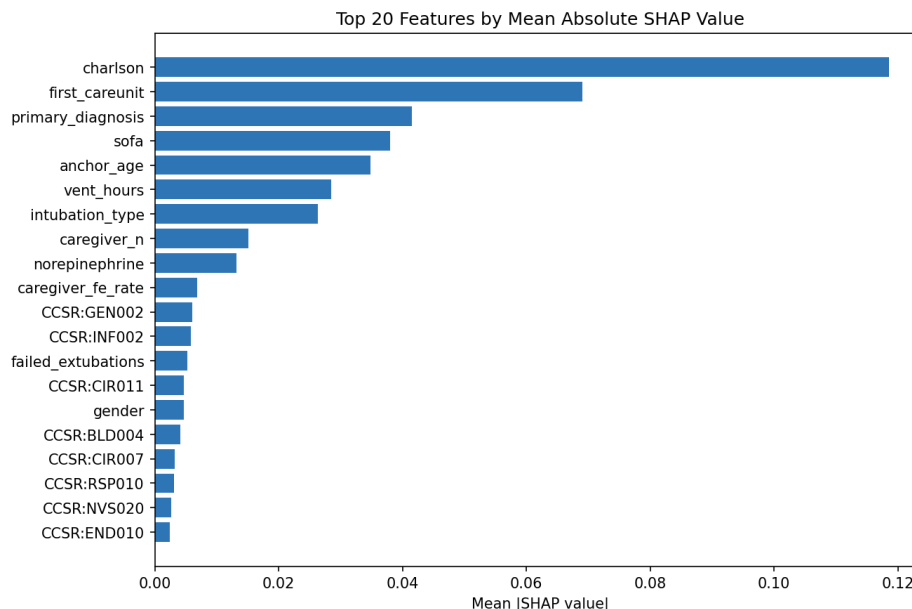


Figure 7: Top 20 features by mean |SHAP value| from the MLP (200 test-set patients, KernelExplainer). Features are ordered by global importance.

The SHAP ranking (Figure 7) is consistent with the R analyses. Charlson comorbidity index is the dominant predictor by a wide margin; `first_careunit` (ICU unit embedding) ranks second — confirming that unit-level case-mix differences are the primary driver of survival heterogeneity. Clinical severity variables (primary diagnosis, SOFA, age, ventilation hours) follow. Caregiver FE rate ranks 10th out of 20, with a negative direction in the SHAP dependence plot: higher FE rate pushes predicted survival downward, consistent with the R PSM result. Its modest rank relative to Charlson and unit reflects the same phenomenon observed in the mixed-effects models — FE rate carries real signal within units but is dominated at the global level by case-mix confounding.

7.3 Double/Debiased ML (DML)

The partially linear DML estimator (Chernozhukov et al. 2018) uses 5-fold cross-fitting with GBM nuisance models to partial out all measured confounders, then regresses the survival residual on the FE rate residual:

Estimator	$\hat{\theta}$	95% CI	p
DML (global, all units)	-0.172	—	0.581
glmmTMB (global mixed effects)	-0.554 (log-odds)	—	0.539

The DML $\hat{\theta}$ is on the probability scale: a one-unit increase in FE rate ($0 \rightarrow 1$, a theoretical maximum) would reduce $P(\text{survival})$ by 17.2 percentage points. Over the observed IQR (≈ 5.5 pp of FE rate), the implied survival difference is < 1 pp — small and not significant ($p = 0.581$). This global null is consistent with the R glmmTMB result ($p = 0.539$) and with the PSM finding that the signal is CVICU-specific: pooling all units, the negative effect in CVICU is diluted by the null in Medical and Surgical/Trauma groups.

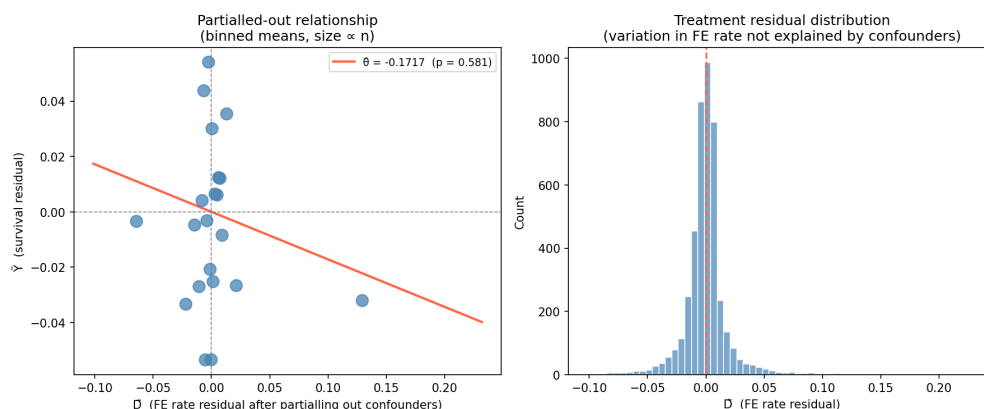


Figure 8: DML partialled-out scatter (left) and treatment residual distribution (right). The negative slope ($\hat{\theta} = -0.172$, $p = 0.581$) is directionally consistent with the R findings but not significant at the global level.

In summary, the deep learning pipeline corroborates the R analysis: the direction of the FE rate effect is consistently negative, the global signal is null once confounders are removed, and the dominant predictors of survival are patient severity and ICU unit rather than caregiver practice.

8 Trajectory Analysis

Andrew’s proposed dynamic risk framework requires that reintubation risk be estimable as a function of time on the ventilator, so that the optimal extubation time can be identified as the point where one more day of waiting produces less than approximately 0.5% reduction in failure probability. We tested whether two standard readiness variables — RSBI and P/F ratio — carry the trajectory information this framework requires.

8.1 Data

RSBI (spontaneous respiratory rate / spontaneous tidal volume in liters) and P/F ratio ($\text{PaO}_2/\text{FiO}_2$) were pulled from MIMIC-IV for all explicit extubation events in the `all_extubations` cohort (6,063 events, 5,505 patients). The outcome is `failed_extubation_flag`: reintubation within 72 hours of this specific extubation event. This is the per-event outcome, not the patient-lifetime reintubation count.

Variable	Events with ≥ 3 measurements	Failed extubations
RSBI	3,551	395 (11.1%)
P/F ratio	3,747	345 (9.2%)

8.2 Individual-level slope analysis

For each extubation event with ≥ 3 measurements in the 72h window, a linear slope was estimated by fitting `lm(measurement ~ hours.before_extubation)`. A negative slope indicates the variable is improving as extubation approaches.

	Succeeded		Failed	
	Mean slope	Mean at extub	Mean slope	Mean at extub
RSBI	-0.782	48.7	-0.290	52.3
P/F ratio	-0.627	256	-0.272	251

Both variables show slopes in the expected direction — succeeded events have more steeply declining (improving) trajectories — but the difference is small relative to within-patient noise.

Logistic regression testing whether slope predicts failure beyond level at extubation:

Predictor	Coefficient	SE	z	p
<i>RSBI model:</i>				
Level at extubation	0.0032	0.0015	2.14	0.032
Slope (72h window)	0.0017	0.0036	0.48	0.634
<i>P/F ratio model:</i>				
Level at extubation	-0.0004	0.0006	-0.67	0.505
Slope (72h window)	0.0015	0.0044	0.35	0.726

RSBI level at extubation is a significant predictor ($p = 0.032$); neither slope is significant ($p = 0.634$ and $p = 0.726$). The trajectory of RSBI or P/F ratio in the 72 hours before extubation carries no independent predictive information beyond the level at the moment of extubation.

8.3 Population-level trajectories

Figure 9 shows hourly mean RSBI and P/F ratio in the 72h before extubation, split by outcome.

8.4 Interpretation

The null slope finding is consistent with two non-exclusive explanations. First, RSBI and P/F ratio may not be sensitive enough variables to detect the trajectory signal — they are measured infrequently (median 4–6 measurements per patient in the 72h window) and are inherently noisy.

Physiological Trajectories Before Extubation by Outcome

Per-extubation-event data (RSBI: 5046 events; P/F: 5327 events).

Hourly bins, $n \geq 30$ per bin. Mean $\pm$ 95% CI. Failed = reintubated within 72h of this specific extubation.

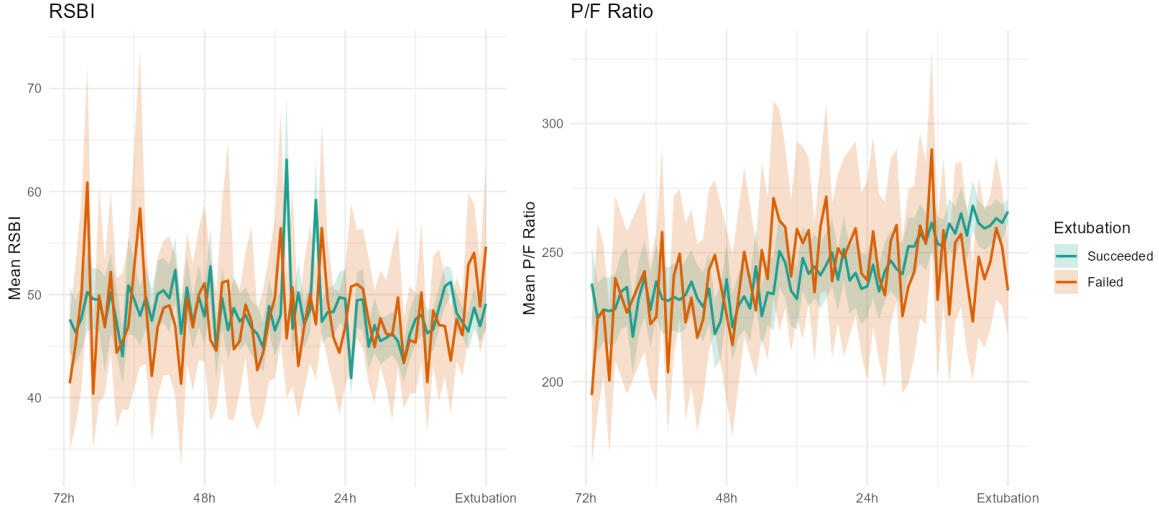


Figure 9: RSBI and P/F ratio in the 72h before extubation by outcome (failed vs. succeeded). Hourly bins, $n \geq 30$ per bin, mean $\pm$ 95% CI. The two groups are largely indistinguishable throughout the window. The zigzag in the failed group reflects genuine measurement noise with approximately 400 failed events distributed across 72 hourly bins.

Second, clinicians may already be conditioning on trajectory implicitly — extubating only after the patient has plateaued at a low RSBI — so that by the time of extubation the trajectory information is spent.

The dynamic risk framework remains theoretically sound. Implementing it prospectively would require variables with higher temporal resolution than are available in retrospective EHR data: continuous pressure support level, FiO_2 weaning trajectory, and spontaneous breathing trial (SBT) outcomes are candidates. Retrospective validation from MIMIC-IV data alone is not feasible with current measurement density.

TODO — SBT outcomes: Spontaneous breathing trial pass/fail is documented in MIMIC-IV chartevents. Whether SBT outcome predicts extubation failure beyond RSBI level is a direct test of the dynamic risk hypothesis and is feasible with existing data.

9 Limitations

1. **Caregiver attribution.** `caregiver_id` records the documenting user. This likely attenuates the signal — a noisy proxy for the performing provider would dilute a real caregiver effect.
2. **Single center.** MIMIC-IV is from Beth Israel Deaconess Medical Center. An eICU parallel analysis (208 ICUs, with caregiver type information) is planned to assess generalizability.
3. **Unmeasured confounding.** PSM controls only for measured covariates. Shift patterns, time of day, attending supervision, and respiratory therapist involvement are not captured in MIMIC-IV.
4. **All-cause 12-month mortality.** Deaths unrelated to the index extubation are counted.

Shorter-term outcomes (hospital mortality, 30-day) are included as secondary endpoints.

5. **Missed extubation opportunities.** Patients extubated too late cannot be identified in retrospective data without knowing the counterfactual. This is the motivation for the dynamic risk analysis (Andrew's Analysis 2).

10 Proposed Next Steps

1. Begin eICU exploratory analysis (208 ICUs, caregiver type information)
2. Agree on citation manager (Zotero?)
3. DML stratified by unit group (to isolate CVICU causal estimate)
4. SBT outcomes analysis (see Section [4.3](#) TODO)